COSC2671 Social Media and Network Analytics

Assignment 2 — Project Worksheet

Group Number	PG Group 18
Member 1	Akash - s4137826
Member 2	Asmi Faisel - s4134240
Member 3	Sreelakshmi Thalodil Sunilkumar - s4146925
Project Title	Who Holds Greater Authority in Online Health Discussions: Licensed Medical Professionals or Wellness Influencers?
Submission Date	25 May 2026

1. Team Project Plan

The team collaborated throughout the project, which ran from May 1 to May 25, 2026, with all members contributing to all phases rather than working in isolation. Weekly meetings were held on Microsoft Teams and organised using Google Calendar. All code and report files were shared using a collaborative workspace, and the final version was sent to the GitHub repository for final edits, version control, and submission preparation.

Week	Dates	Tasks	Owner	Status
Week 1	1–7 May	Topic finalisation, proposal submission, YouTube API setup, data collection (doctor queries), GitHub repository setup, team roles allocation	All	Done
Week 2	8–14 May	Influencer data collection, data merging and deduplication, text cleaning and preprocessing, video labelling, exploratory data analysis	Akash, Asmi	Done
Week 3	15–19 May	Tokenisation, unigram/bigram analysis, VADER sentiment analysis, LDA topic modelling, network construction, centrality analysis	Asmi, Sreelakshmi	Done
Week 3-4	18–21 May	Louvain community detection, network visualisations, hub-and-spoke graph, degree distribution	Akash, Sreelakshmi	Done
Week 4	20–22 May	Report writing, figure insertion, presentation preparation, final review and submission	All	Done

2. Weekly Timesheets

Each team member kept a weekly timesheet that tracked work completed, hours spent, and tools utilised. All timesheets are shown below.

2.1 Akash (s4137826)

Week	Date	Task	Tool	Hours
Week 1	1 May	Research question formulation and looked into Youtbe comments	MS Teams	2.0
Week 1	3 May	API quota testing and comment collection planning	Jupyter	2.5
Week 1	5 May	Topic proposal drafting and peer review	Google Calendar	3.0
Week 1	7 May	Task allocation and timeline review	MS Teams	1.5
Week 2	8 May	Text cleaning pipeline	Jupyter	1.5
Week 2	10 May	ASCII filter, short comment removal, deduplication	Python	3.0

Week 2	12 May	TweetTokenizer implementation and custom stopword list	NLTK	2.0
Week 2	13 May	Boxplot visualisation (before/after preprocessing)	Matplotlib	2.5
Week 3	15 May	Unigram frequency analysis	Python	2.0
Week 3	16 May	Bigram frequency analysis	NLTK	3.0
Week 3	17 May	Word frequency bar charts	Matplotlib	2.0
Week 3	18 May	VADER sentiment analysis implementation	NLTK	2.5
Week 3	19 May	Sentiment visualisations	Matplotlib	3.0
Week 4	20 May	LDA topic modelling	scikit-learn	2.5
Week 4	21 May	Word cloud generation	WordCloud	2.0
Week 4	22 May	Text analysis sections of report writing	MS Teams	2.5
Week 4	23 May	Presentation slides — NLP section	MS Teams	2.0
Week 4	25 May	Final report review and proofreading	GitHub	1.5
		Total Hours		**41.0**

2.2 Asmi Faisel (s4134240)

Week	Date	Task	Tool	Hours
Week 1	1 May	Project scope research and competitor analysis review	MS Teams	2.5
Week 1	4 May	Data collection planning and query selection and filtering	Jupyter	3.0
Week 1	5 May	Topic proposal review and submission coordination	MS Teams	2.0
Week 1	7 May	GitHub onboarding and repository familiarisation	GitHub	1.5
Week 2	9 May	Dataset merging validation and label distribution check	Python	3.0
Week 2	11 May	Exploratory analysis	Matplotlib	3.0
Week 2	12 May	Comment distribution histogram (initial exploration)	Matplotlib	2.5
Week 2	14 May	Data quality review	Jupyter	2.5
Week 3	15 May	Eigenvector and clustering coefficient computation	NetworkX	3.5
Week 3	16 May	Degree distribution histogram	Matplotlib	2.5
Week 3	17 May	Top 10 user's centrality bar chart	Matplotlib	3.0
Week 3	18 May	Report writing (introduction and data sections)	MS Teams	4.0
Week 3	19 May	Report writing (preprocessing and labelling sections)	MS Teams	3.5
Week 4	20 May	Report writing (network analysis section)	MS Teams	4.0
Week 4	21 May	Report writing (discussion and conclusion sections)	MS Teams	3.5
Week 4	22 May	Final figure insertion and table formatting in report	MS Word	3.0
Week 4	23 May	Presentation slides (introduction and data sections)	MS Teams	2.5
Week 4	25 May	Final submission preparation and file naming	GitHub	2.0
		Total Hours		**51.5**

2.3 Sreelakshmi Thalodil Sunilkumar (s4146925)

Week	Date	Task	Tool	Hours
Week 1	1 May	Formulating research questions and brainstorming topics	MS Teams	3.0
Week 1	2 May	Setting up and testing YouTube API v3 authentication	GitHub, Python	2.5
Week 1	3 May	Data collecting using doctor enquiries	Jupyter	2.5
Week 1	5 May	Topic proposal writing and submission	MS Teams	1.5
Week 1	7 May	Branch structure, README, and GitHub repository setup	GitHub	4.0
Week 2	8 May	Influencer query data collection	Jupyter	3.5
Week 2	9 May	Video ID-based dataset merging and deduplication	Python	3.5
Week 2	11 May	Logic for video tagging (keyword scoring method)	Jupyter	2.5
Week 2	13 May	Charts of comment distribution and exploratory data analysis	Matplotlib	3.5
Week 3	15 May	Building a directed graph from reply data using NetworkX	Jupyter	3.0
Week 3	16 May	Graph cleaning and self-loop elimination	Python	3.5
Week 3	18 May	Calculation of degree and betweenness centrality	NetworkX	3.5
Week 3	19 May	Hub-and-spoke network visualisation with colour coding	Matplotlib	2.5
Week 4	20 May	Louvain community detection implementation	python-louvain	4.0
Week 4	21 May	Community-coloured network visualisation	Matplotlib	2.0
Week 4	22 May	Network analysis section of report writing	MS Teams	4.0
Week 4	23 May	Presentation slides network analysis section	MS Teams	2.5
Week 4	25 May	Final review, code cleanup, GitHub push	GitHub	2.0
		Total Hours		**53.5**

3. Work Breakdown Summary

Throughout the project's life cycle, the team collaborated. While some individuals assumed major responsibility for specific areas, all members actively participated in debates, code reviews, and writing. The final version of all code and the report has been uploaded to the shared GitHub repository for final edits and submission.

Task Area	Akash s4137826	Asmi Faisel s4134240	Sreelakshmi s4146925
Data Collection & API	Lead	Support	Lead
Data Preprocessing & Cleaning	Lead	Support	Support
Video Labelling	Support	Lead	Lead
Text Analysis (Unigram/Bigram)	Lead	Support	Support
Sentiment Analysis (VADER)	Lead	Support	Support
Topic Modelling (LDA)	Lead	Support	Support

Network Construction	Support	Support	Lead
Centrality Analysis	Support	Lead	Lead
Community Detection (Louvain)	Support	Support	Lead
Network Visualisation	Support	Lead	Lead
Report Writing	Support	Lead	Lead
Presentation Preparation	Equal	Equal	Equal
Total Hours	**41.0 hrs**	**53.5 hrs**	**51.5 hrs**

4. Individual Self-Reflection

Each team member reflects on their individual performance, team dynamics, and lessons learned throughout the project. The combined self-reflection section is limited to 1 A4 page in total at font size 12.

Akash (s4137826)

Went well: The NLP pipeline came together well — from text cleaning through to VADER sentiment analysis and LDA topic modelling. The TweetTokenizer handled informal social media language effectively and the custom stopword list improved token quality significantly. Working collaboratively with the team meant that when I was stuck on something, others could help and vice versa.

Challenging: Selecting the optimal number of LDA topics required group discussion and experimentation as there was no single objectively correct answer. Non-English Romanised comments in the dataset affected topic model output unexpectedly and required acknowledgement as a dataset limitation.

Do differently: Instead of relying on manual examination, I would use coherence score evaluation to select LDA topics from the start. I would also begin the text analysis earlier in the project to allow for more time to iterate on the results prior to the report deadline.

Asmi Faisel (s4134240)

Went well: The report writing was well-coordinated across the team, and it followed a logical structure from data collecting to conclusion. The computation of eigenvectors and clustering coefficients yielded meaningful findings. Using Microsoft Teams for communication and Google Calendar for scheduling kept everyone on track throughout the project.

Challenging: Coordinating report writing while also conducting the centrality analysis was difficult. Maintaining consistency in figure numbering and table references across sections produced by various members necessitated diligent examination.

Do differently: I would create a shared report template from the start so that all members may edit together rather than merging different versions at the end. I would also set up a mid-project rubric review session to discover holes earlier.

Sreelakshmi Thalodil Sunilkumar (s4146925)

Went well: The data collecting and network creation components functioned properly. Building the directed reply graph with NetworkX and applying Louvain community detection yielded clear and interpretable findings. The final GitHub push provided the team with a clear shared version to review and amend before submission, which was really useful for discovering last-minute errors.

Challenging Initial network graphs revealed self-loops due to YouTube reply metadata, necessitating extra debugging before meaningful visualisations could be produced. Coordinating the final GitHub push and ensuring all files were properly titled under time constraints was also difficult.**:**

Do differently: I would validate the reply graph structure earlier in the project to catch the self-loop problem sooner. I would also establish the GitHub repository naming conventions and folder structure at the outset to avoid last-minute reorganisation before submission.